AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

Assignment Title:	WBS, Effort Estimation, Scheduling		
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Course Title:	Software Engineering		
Course Code:	00491	Section:	**T**
Semester:	Spring 2024-25	Course Teacher:	**TONNY SHEKHA KAR**

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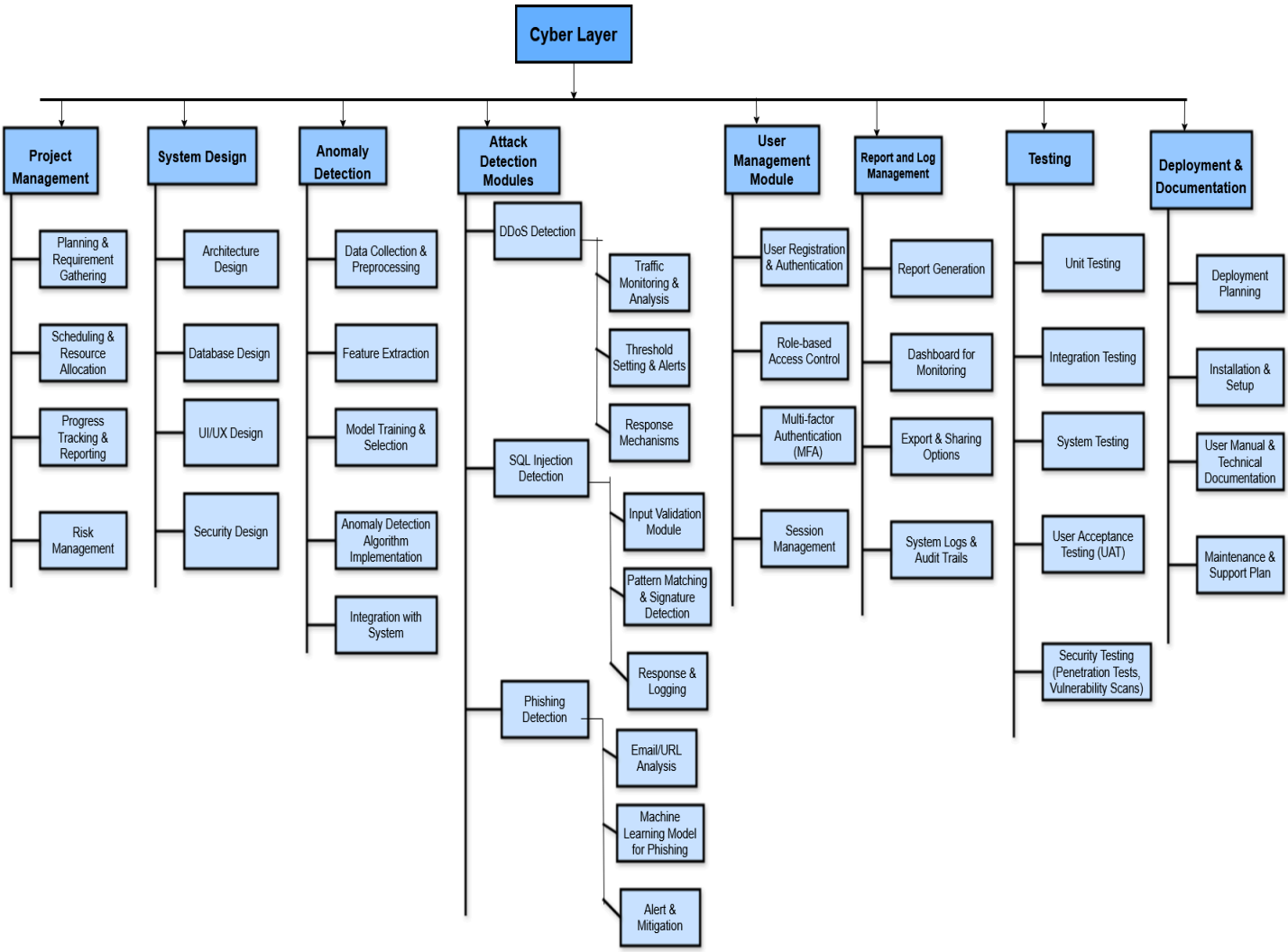
Faculty use only		
FACULTY COMMENTS	**Marks Obtained**	
	Total Marks	

WBS:


Cyber Layer
Project Management
Planning & Requirement Gathering
Scheduling & Resource Allocation
Progress Tracking & Reporting
Risk Management
System Design
Architecture Design
Database Design
UI/UX Design
Security Design
Anomaly Detection
Data Collection & Preprocessing
Feature Extraction
Model Training & Selection
Anomaly Detection Algorithm Implementation
Integration with System
Attack Detection Modules
DDoS Detection
Traffic Monitoring & Analysis
Threshold Setting & Alerts
Response Mechanisms
SQL Injection Detection
Input Validation Module
Pattern Matching & Signature Detection
Response & Logging
Phishing Detection
Email/URL Analysis
Machine Learning Model for Phishing
Alert & Mitigation
User Management Module
User Registration & Authentication
Role-based Access Control
Multi-factor Authentication (MFA)
Session Management
Report and Log Management
Report Generation
Dashboard for Monitoring
Export & Sharing Options
System Logs & Audit Trails
Testing
Unit Testing
Integration Testing
System Testing
User Acceptance Testing (UAT)
Security Testing (Penetration Tests, Vulnerability Scans)
Deployment & Documentation
Deployment Planning
Installation & Setup
User Manual & Technical Documentation
Maintenance & Support Plan

Project Type:

CyberLayer is an embedded software system based on the COCOMO model because it requires intricate integration of software and hardware to perform real-time anomaly and attack detection. It has high performance requirements, is able to interface with network hardware or sensors, and involves close integration of behavioral and signature-based attack detection modules. Its security-critical use and system complexity and the timing sensitivity characterize it nearly to perfection within the embedded systems' parameters.

Effort Estimation:

Software Project Type: Embedded

Software Project Type	Coefficient	(P)	SLOC-dependent Coefficient (T)
Organic	2.4	1.05	0.38
Semi-detached	3.0	1.12	0.35
Embedded	**3.6**	**1.20**	**0.32**

Constructive Cost Model (COCOMO):

Effort, PM = Coefficient<Effort Factor> * $(SLOC/1000)^P$

$= 3.5 * (7000/1000)^{1.2}$

= 37.66 person-months

= 38 person-months

Development time, DM = $2.50 * (PM)^T$

$= 2.50 * (37.66)^{0.32}$

= 7.95 months ≈ 8 months

Required number of people ST = PM/DM

= 37.66 /8

= 4.74

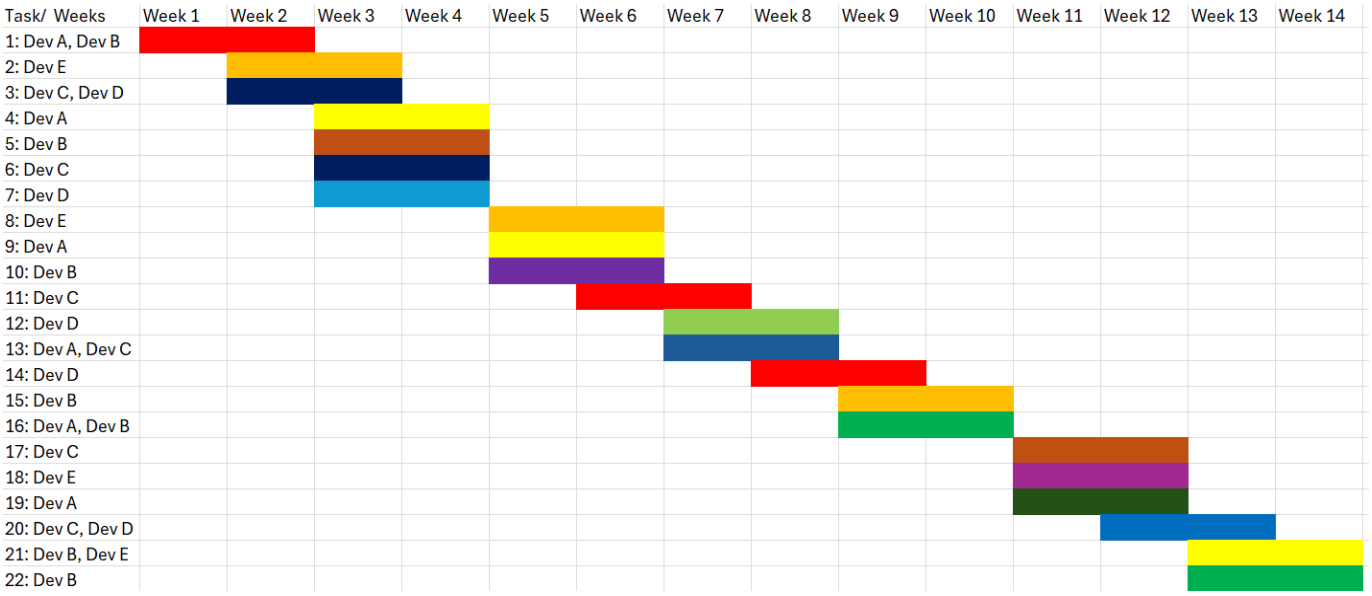
≈ 5 people

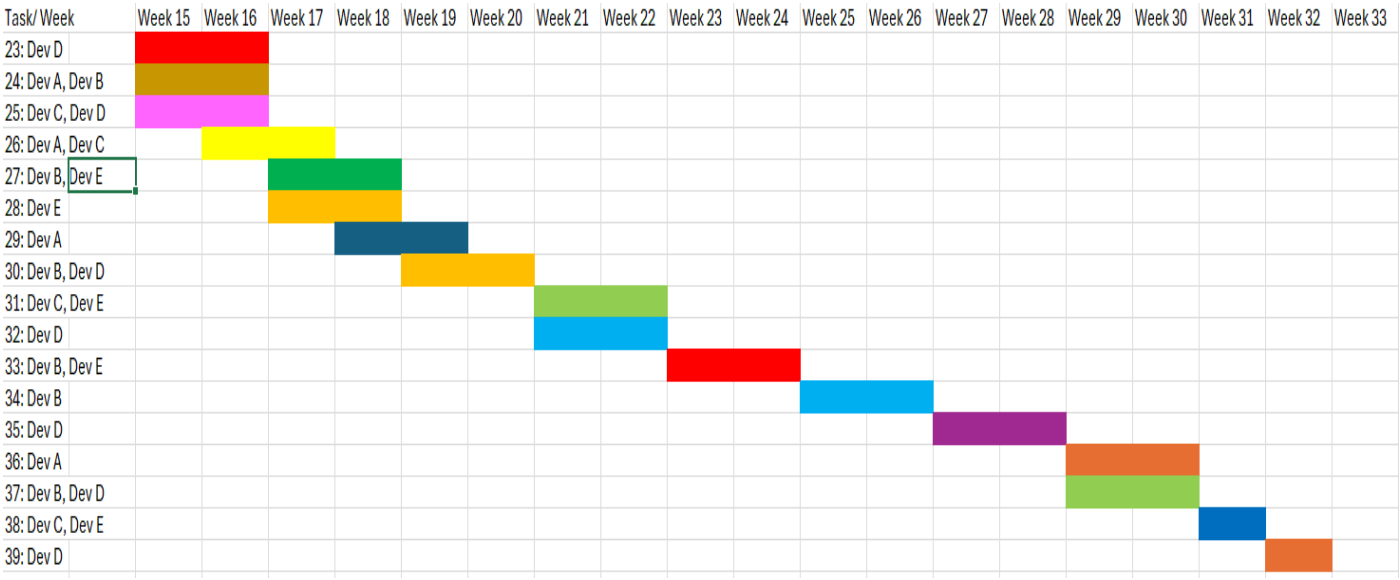
$$\text{Weeks} = DM \times 4$$

$$= 8 \times 4$$

$$= 32$$

Timeline Chart (I)





Task Activity:

1. Planning & Requirement Gathering
2. Architecture Design
3. Scheduling & Resource Allocation
4. Progress Tracking & Reporting
5. Database Design (ER Diagram, Schema)
6. UI/UX Design (Wireframes, Mockups)
7. Risk Management
8. Security Design (MFA, Encryption, Access Control)
9. Data Collection & Preprocessing
10. User Registration & Authentication
11. Feature Extraction
12. Role-based Access Control
13. Report Generation
14. Model Training & Selection
15. Multi-factor Authentication (MFA)
16. Dashboard for Monitoring
17. Anomaly Detection Algorithm Implementation
18. Session Management
19. Export & Sharing Options
20. Integration with System
21. System Logs & Audit Trails
22. Traffic Monitoring & Analysis

23. Threshold Setting & Alerts

24. Input Validation Module

25. Email/URL Analysis

26. Response Mechanisms

27. Pattern Matching & Signature Detection

28. Machine Learning Model for Phishing

29. Response & Logging

30. Alert & Mitigation

31. Unit Testing

32. Integration Testing

33. System Testing

34. User Acceptance Testing (UAT)

35. Security Testing (Penetration Tests, Vulnerability Scans)

36. Deployment Planning

37. Installation & Setup

38. User Manual & Technical Documentation

39. Maintenance & Support Plan

Time chart (II)

WBS ID	Work Tasks	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
1	Planning & Requirement Gathering						
2.1	Architecture Design						
1.2	Scheduling & Resource Allocation.						
	Milestone: Project goals defined and initial architecture drafted.						
1.3	Progress Tracking & Reporting						
2.2	Database Design						
2.3	UI/UX Design						
1.4	Risk Management.						
	Milestone: Approved UI design and risk management plan finalized.						
2.4	2.4 Security Design						
3.1	3.1 Data Collection & Preprocessing						
5.1	5.1 User Registration & Authentication						
	Milestone: Fully functional authentication module						

WBS ID	Work Tasks	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12	Week 13	Week 14
3.2	Feature Extraction								
5.2	Role-based Access Control								
6.1	Report Generation.								
	Milestone: Initial report generation module operational.								
3.3	Model Training & Selection								
5.3	Multi-factor Authentication								
6.2	Dashboard for Monitoring								
	Milestone: Monitoring dashboard ready for testing.								
3.4	3.4 Anomaly Detection Algorithm Implementation								
5.4	5.4 Session Management								
6.3	6.3 Export & Sharing Options.								
	Milestone: Session management module completed.								
3.5	3.5 Integration with System								
6.4	6.4 System Logs & Audit Trails								
4.1.1	4.1.1 Traffic Monitoring & Analysis								
	Milestone: Traffic monitoring module functional.								

WBS ID	Work Tasks	Week 15	Week 16	Week 17	Week 18	Week 19	Week 20	Week 21	Week 22	Week 23	Week 24	W
4.1.2	4.1.2 Threshold Setting & Alerts											
4.2.1	4.2.1 Input Validation Module											
	Milestone: Thresholds tested and input validation completed											
4.1.3	4.1.3 Response Mechanisms											
4.2.2	4.2.2 Pattern Matching & Signature Detection											
4.3.2	4.3.2 ML Model for Phishing											
	Milestone: Response mechanisms and phishing model developed.											
4.2.3	Response & Logging											
4.3.3	Alert & Mitigation											
	Milestone: Response logging finalized.											
7.1	7.1 Unit Testing											
	Milestone: Unit testing completed for all modules.											
7.2	7.2 Integration Testing											
	Milestone: Integration testing completed successfully.											
7.3	7.3 System Testing											
	Milestone: System testing completed.											

WBS ID	Work Tasks	Week 25	Week 26	Week 27	Week 28	Week 29	Week 30	Week 31	Week 32
7.4	User Acceptance Testing (UAT)								
	Milestone: UAT feedback								
7.5	Security Testing (Penetration Tests, Vulnerability Scans).								
	Milestone: Security issues resolved or mitigated.								
8.1	Deployment Planning.								
	Milestone: Deployment plan finalized.								
8.2	Installation & Setup.								
	Milestone: System installed and setup completed.								
8.3	User Manual & Technical Documentation.								
	Milestone: Documentation completed and reviewed.								
8.4	Maintenance & Support Plan.								
	Milestone: Maintenance and support plan finalized.								